

# **The Aggregate Impact of Policy Uncertainty in the Green Energy Transition**

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# Motivation

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*“Transitioning the global economy from fossil fuels to sustainable sources of energy requires the largest reallocation of capital in human history and requires spending of \$3.5T a year”*

(Institute for Climate and Sustainability at Harvard University)

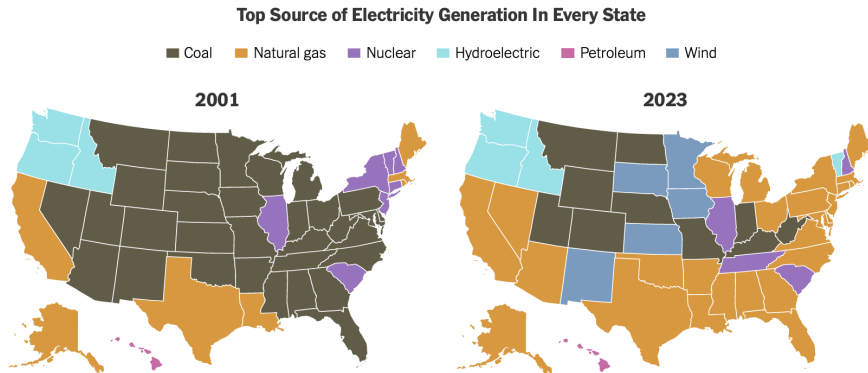
- Electricity demand is projected to grow 0.9–1.6% per year through 2050 (EIA AEO 2025)
- Subsidies play a key role in the transition: by 2024 the U.S. spent \$221B on low-carbon electricity subsidies (IEA)
- Investors make long-run decisions without knowing whether the subsidy will be available

▸ Cost per MWh

▸ Capital costs

▸ Generation mix

# The Transition Is Heterogeneous



Source: U.S. Energy Information Administration

# This Paper

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**Research question:** What are the macroeconomic and environmental costs of renewable policy uncertainty?

- **Model:** dynamic spatial general-equilibrium model with the U.S. electricity sector
  - 13 regions trading within 3 interconnections; irreversible wind, solar, and gas capital
  - The subsidy is stochastic, calibrated to switches in party control
- **Budget-neutral counterfactuals:** permanent subsidy vs. an uncertain subsidy

# Renewable Energy Trade-off: When to Invest?

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## Force 1 — Wait for falling costs

Solar capital costs have fallen 83% since 2010; wind 55%.

Building 1 GW of solar in 2010 instead of 2009 saved  $\approx$  **\$220M** in capital costs: one year of delay.

⇒ **Wait**: every year of delay means cheaper capital.

## Force 2 — Act now to reduce emissions

Each year of delay means one more year of fossil generation.

1 GW of solar displaces  $\approx$  2.2 TWh of gas a year  $\approx$  0.9 Mt CO<sub>2</sub>  $\approx$  **\$170M in climate damages** (SCC \$190/t).

⇒ **Act now**: delay has climate costs, which firms do not pay.

- Subsidy permanently on ⇒ same trade-off, just lower costs
- Subsidy on and off ⇒ **wait** for the subsidy to return; build more while it lasts

# Main Results

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Relative to a budget-neutral uncertain subsidy, a permanent subsidy:

- Adds **13%** (137 GW) more renewables and **4.4%** (37 GW) less gas
- Lowers cumulative CO<sub>2</sub> by **3.2%** (2.29 Gt)
- Raises welfare by **\$208 billion** in present value

Two further results:

- Effects are largest where federal support changes renewable investment and existing fossil capacity can fill the generation gap during lapses (the Midwest)
- **Perfect foresight does not eliminate the value of permanence:** it is \$245B with a known 6-year lapse and \$365B with a known 17-year lapse

# Related Literature

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- **Macroeconomics of the clean-energy transition**

- Acemoglu et al. (2012); **Arkolakis & Walsh (2023)**; **Casey et al. (2026)**; Conte et al. (2026); Fried (2018); Golosov et al. (2014): energy as a clean/dirty aggregate, policy taken as given
- **This paper:** a detailed electricity sector (merit order, peak demand) and a *stochastic* policy

- **Policy uncertainty and investment**

- Barradale (2010); **Chen (2025)**; **Fried, Novan & Peterman (2022)**; Lemoine (2017)
- **This paper:** *spatial* general-equilibrium allows to quantify policy spillovers

- **Renewable subsidies and electricity markets**

- Langer & Lemoine (2022), Aldy et al. (2023), Gowrisankaran et al. (2025)
- **This paper:** the electricity sector embedded in general equilibrium

# Empirical Evidence

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# Institutional Background

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- Two credits, both locked in at the start of construction
  - Wind: Production Tax Credit, a payment per MWh
  - Solar: Investment Tax Credit, a share of capital cost
- **Historically cross-party:** lapses were short legislative delays 
  - Wind credit lapsed four times; each time renewed retroactively 
  - Survived unified Republican control in 2003–2006 and 2017–2018
- **Now partisan**
  - Republican support for the credit bills: 75% (1992) → 0%
  - Expanding clean energy: “very important” for 73% of Democrats, 39% of Republicans (University of Chicago Energy Policy Institute survey, 2025)

# Fact 1: Much of U.S. Wind Needs the Subsidy

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**Exercise:** compare each wind plant's *realized* capacity factor to the CF it needs to *break even*, with and without the credit.

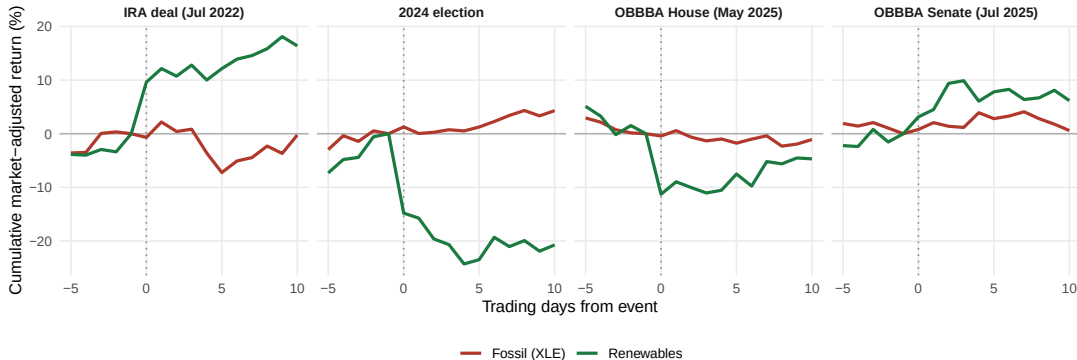
## Inputs, all from data:

- Capacity factors: plant-level EIA-923 net generation  $\div$  EIA-860 nameplate (2021) — 1,203 plants, 129 GW
- Wholesale price: \$50/MWh, the 2010–2024 on-peak average across 8 EIA/ICE hubs (on-peak  $\Rightarrow$  lower bound)
- Breakeven CF (NREL ATB engineering cost): **0.35** wholesale, **0.23** with the PTC

## Of 129 GW of operating wind (2021):

- 48% viable without the credit
- **35% viable only with the credit** (60% at the \$40/MWh pre-2021 price)
- 17% below even the subsidized breakeven

## Fact 2: The Recent Reversals Were Unforeseen



- Inflation Reduction Act (IRA) extended subsidies for 10 years
- One Big Beautiful Bill Act (OBBBA) repealed them
- Only renewables move, only on the news day  $\Rightarrow$  reversals were unforeseen

# Model

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# Model Overview

13 EIA regions, trade *within* 3 interconnections; annual, infinite horizon from 2010

| Agent               | Level    | Problem                        | Key choices                   | State              |
|---------------------|----------|--------------------------------|-------------------------------|--------------------|
| Household-generator | Regional | Utility maximization (dynamic) | $C, E^{HH}, S, I^S, I^W, I^G$ | $A, \mathbf{K}, s$ |
| Final-good firm     | Regional | Profit maximization (static)   | $K_P, L, E^{firm}$            | —                  |
| Government          | National | Budget balance                 | $\varsigma_t, T_t^{nat}$      | —                  |

- Regions differ in wind/solar resource, demand peaks, initial capital, hydro/nuclear
- Exogenous paths: electricity demand, technology costs, fuel prices, coal retirement [▶ Paths](#)
- **Policy:** the only source of uncertainty;  $s_t \in \{0, 1\}$  follows a Markov chain; agents know the probabilities, not the timing

[▶ Map](#)[▶ Trade](#)[▶ Capacity factors](#)

# Household-Generator's Problem

Representative household in region  $j$ : consumes, supplies labor inelastically, saves in a national market, **owns and invests in generation capital**.

$$\max \mathbb{E}_0 \sum_t \beta^t \log U(C_{jt}, E_{jt}^{HH}), \quad U(C, E) = \left[ C^{\frac{\sigma-1}{\sigma}} + \bar{\alpha}_h^{1/\sigma} E^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

**Budget:**

$$\underbrace{C_{jt} + p_{jt}^E E_{jt}^{HH}}_{\text{consumption}} + \underbrace{\sum_k \tilde{p}_t^k I_{jt}^k + \sum_k \Psi^k(I_{jt}^k, K_{jt}^k)}_{\text{generation investment}} + \underbrace{S_{jt}}_{\text{saving}} \\ = \underbrace{(1 - \varsigma_t) w_{jt} L_{jt}}_{\text{after-tax wages}} + \underbrace{r_t^K A_{jt}}_{\text{capital income}} + \underbrace{\Pi_{jt}^E}_{\text{electricity profit}} + \underbrace{\mathcal{R}_{jt} + N_j T_t^{\text{nat}}}_{\text{transfers}}$$

- Solar subsidy lowers the effective price of solar capital:  $\tilde{p}_t^S = (1 - \phi^{ITC} \chi_t^S) p_t^S$ ; wind subsidy is paid per MWh inside  $\Pi^E$
- Adjustment cost  $\Psi^k = \frac{\psi^k}{2} (I/K)^2 K$ ; rising wind site cost  $c_j^W(K) = \tilde{p}_t^W [1 + (K/\bar{K}_j)^\eta]$
- Irreversible:  $I_{jt}^k \geq 0$ ,  $K_{j,t+1}^k = (1 - \delta^k) K_{jt}^k + I_{jt}^k$

# Production and Electricity Demand

**Final-good firm** in region  $j$ : CES in a capital–labor bundle and electricity

$$Y_{jt} = Z_{jt} \left[ (1 - \gamma_E)^{1/\sigma} (K_{P,jt}^{\hat{\alpha}_K} L_{jt}^{\hat{\alpha}_L})^{\frac{\sigma-1}{\sigma}} + \gamma_E^{1/\sigma} (E_{jt}^{firm})^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

**Electricity demand** (firms and households, elasticity  $\sigma = 0.3$ ):

$$E_{jt}^{firm} = \bar{\alpha}_E Y_{jt} (p_{jt}^E)^{-\sigma}, \quad E_{jt}^{HH} = \bar{\alpha}_h C_{jt} (p_{jt}^E)^{-\sigma}$$

- Electricity cost share 2% (BEA); annual demand  $\bar{E}_{jt} = E_{jt}^{firm} + E_{jt}^{HH}$  sets the scale of hourly demand in the merit order
- Firms rent production capital in a national market,  $\sum_j K_{P,jt} = \sum_j A_{jt}$  at  $r_t^K$ ; labor is supplied inelastically

# Investment Decision

**Bellman equation for generation capital**, discounted with the stochastic discount factor  $M_{t,t+1}$ :

$$V_t(\mathbf{K}_{jt}, s_t) = \max_{I^S, I^W, I^G \geq 0} \left\{ \underbrace{\Pi_{jt}^E(\mathbf{K}_{jt}, s_t)}_{\text{electricity profit}} - \underbrace{[\tilde{p}_t^S I^S + \Xi_{jt}^W(I^W) + p_t^G I^G]}_{\text{investment cost}} - \underbrace{\sum_k \Psi^k}_{\text{adjustment cost}} + \underbrace{\mathbb{E}[M_{t,t+1} V_{t+1}(\mathbf{K}_{j,t+1}, s_{t+1}) | s_t]}_{\text{continuation value under policy uncertainty}} \right\}$$

Only wind started while the subsidy is on joins the eligible stock  $K^{W,PTC}$  and earns the subsidy for 10 years.

► Vintage rule

**Wind first-order condition:**

$$\underbrace{c_j^W(K_{jt}^W) + \psi_I^W}_{\text{marginal cost}} = \underbrace{\mathbb{E}[Mq_{t+1}^W | s_t]}_{\text{energy value}} + \underbrace{\mathbf{1}[s_t = 1] \cdot \mathbb{E}[Mq_{t+1}^{W,PTC} | s_t]}_{\text{subsidy value}}$$

- In a lapse the subsidy term is zero: wind vs. gas compete on own costs
- If a lapse is likely, new wind may never earn the subsidy  $\Rightarrow$  gas looks safer

► Why a separate Bellman? Fisher separation & GE

# Electricity Sector

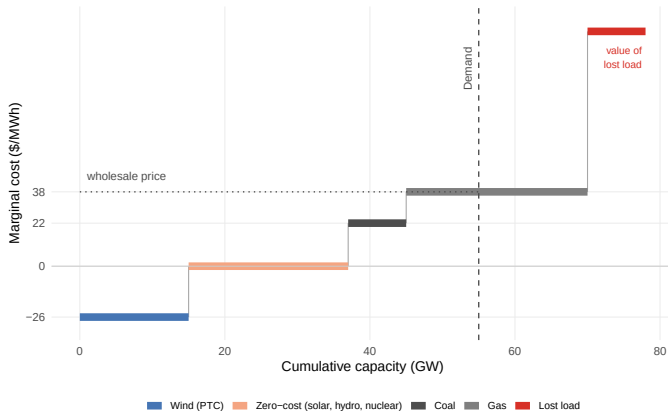
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- **Six technologies:** regions invest in wind, solar, and gas; hydro and nuclear are fixed; coal retires on an exogenous path
- **Capacity  $\neq$  generation:** regions choose capacity; plants are dispatched hour by hour, cheapest first (merit order)
- **Hourly demand peaks** (Pareto): the price is set by the last plant dispatched, in closed form
  - renewables lower the price they earn  $\Rightarrow$  without the subsidy, wind stops at an interior level
  - wind and solar cannot cover the peaks  $\Rightarrow$  gas is the backup
- **Emissions:**  $\mathcal{E}_t = \sum_j (\xi^G gen_{jt}^G + \xi^C gen_{jt}^C)$
- **Trade** within 3 interconnections: local price = hub price minus a congestion discount ▶ Trade

▶ vs. standard macro-energy models

▶ Evidence from data

# Electricity Prices: Merit-Order Dispatch



More renewables push gas out of the margin  $\Rightarrow$  lower prices

► Duck curve

► Texas

► Neg. prices

# Electricity Prices and Profit

**Expected price** (closed form, Pareto hourly demand):

$$\bar{p}_{jt}^E = mc_{jt}^{lo} \left( \frac{x_{jt}^{\min}}{x_{jt}^{d2}} \right)^{\theta_j} + (mc_{jt}^{hi} - mc_{jt}^{lo}) \left( \frac{x_{jt}^{\min}}{x_{jt}^{d3}} \right)^{\theta_j} + (V_j - mc_{jt}^{hi}) \left( \frac{x_{jt}^{\min}}{Q_{jt}} \right)^{\theta_j}$$

- Each term: the probability that demand reaches the next plant type, times the price step
- More renewables raise the thresholds  $x^{d2}, x^{d3} \Rightarrow$  lower expected price

**Electricity-sector profit:**

$$\Pi_{jt}^E = \underbrace{p_{jt}^E G_{jt}}_{\text{sales}} + \underbrace{\phi^{PTC} CF_j^W 8760 K_{jt}^{W,PTC}}_{\text{wind subsidy, 10 years}} - \underbrace{mc_t^G gen_{jt}^G - mc_t^C gen_{jt}^C}_{\text{fuel costs}}$$

- Subsidy paid per MWh, only on wind started while the subsidy is on ( $K^{W,PTC}$ )

# Equilibrium

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**Within each period:** given capital  $\mathbf{K}_{jt}$  and policy state  $s_t$ , prices clear electricity markets

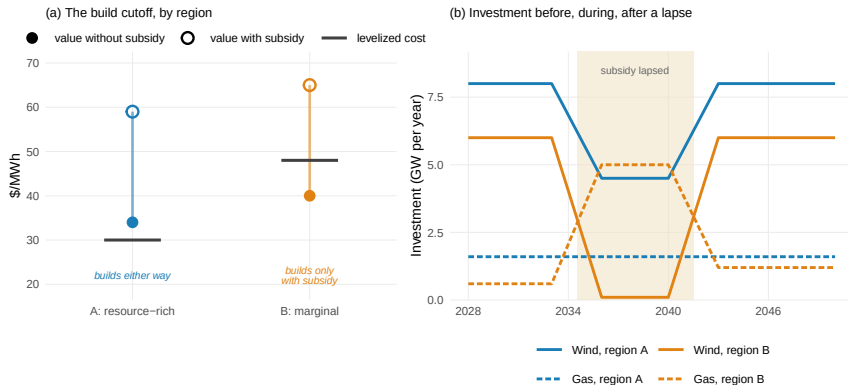
- Each region dispatches by merit order; price = cost of the last plant dispatched
- Household and firm electricity demand optimal at those prices
- Demand = supply within each interconnection

**Over time:** a recursive competitive equilibrium from 2010 to the stochastic steady state

- Investment solves the household-generator's problem, with irreversibility and the subsidy vintage
- Stochastic discount factor  $M_{t,t+1} = \beta u_C(t+1)/u_C(t)$  from household consumption
- National capital market, regional labor markets, and goods market clear
- Government budget balances: labor tax pays subsidies and a lump-sum transfer
- Policy state follows the Markov chain  $\Pi$

Each counterfactual is this equilibrium under a different forward  $\Pi$ .

# Mechanism: Which Regions Build



Invest in wind if the value of a new MWh plus the subsidy covers its levelized cost

► Midwest in the full model

# Mechanism: Option Value of Waiting

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Marginal region during a lapse; a marginal project breaks even at the wholesale price.

## Lapse is permanent

- Nothing to wait for
- Invest now if the profit is positive

## Subsidy may return

- A project started in a lapse *never* gets the subsidy
- Subsidy worth  $\approx$ **\$14/MWh** levelized (\$26 for 10 yrs)
- Even a return probability of  $\approx 0.06/\text{yr}$  makes **waiting** optimal

The *chance that the subsidy comes back*, not the subsidy level, delays the project.

## While renewable investment is delayed, fossil generation fills the gap:

- existing coal and gas plants run more
- some additional gas capacity is built as demand grows
- new gas is irreversible: once built, it remains in the generation stock until it depreciates

# Welfare

Two components, reported separately:

$$W^C = \underbrace{\sum_t \beta^t \sum_j N_j \log \mathcal{U}(C_{jt}/N_j, E_{jt}^{HH}/N_j)}_{\text{consumption and electricity}} \quad W^E = \underbrace{\sum_t \left( \frac{1}{1+r_c} \right)^{t-2010} \text{SCC}_t E_t}_{\text{climate damage}}$$

- Consumption discounted at the model's rate ( $\beta$ ); reported as consumption-equivalent % and in dollars
- Climate damage: EPA year-specific SCC (\$190/t), discounted at the EPA's social rate  $r_c = 2\%$
- **Value of permanence** = avoided climate damage – consumption cost, permanent vs. uncertain subsidy

# Calibration

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# Calibration

| Parameter                             | Symbol                       | Value                           | Source / target                   |
|---------------------------------------|------------------------------|---------------------------------|-----------------------------------|
| <i>A. Set externally</i>              |                              |                                 |                                   |
| Discount factor                       | $\beta$                      | 0.9615                          | $\rho \approx 4\%$                |
| Demand elasticity                     | $\sigma$                     | 0.3                             | literature: 0.2–0.4               |
| Capital-cost paths                    | $p_t^S, p_t^W, p_t^G$        | declining                       | NREL ATB 2024, LBNL/EIA           |
| Electricity demand path               |                              | rising 0.9–1.6%/yr              | EIA AEO 2025                      |
| Capacity factors                      | $CF_j^k$                     | wind 0.11–0.36, solar 0.17–0.27 | EIA-923/860                       |
| Subsidy rates                         | $\phi^{PTC}, \phi^{ITC}$     | \$26/MWh, 30%                   | tax code §45, §48                 |
| Emission factors                      | $\xi^G, \xi^C$               | 0.40, 0.95 t/MWh                | EPA eGRID                         |
| <i>B. Matched internally</i>          |                              |                                 |                                   |
| Wind, solar adjustment cost           | $\bar{\psi}^W, \bar{\psi}^S$ | $2 \times 10^5$                 | 2010–2024 capacity growth         |
| Gas adjustment cost                   | $\bar{\psi}^G$               | $10^7$                          | combined-cycle builds more slowly |
| Wind site-cost scale                  | $\kappa^W$                   | 0.08                            | 2024 wind: 143 GW                 |
| Solar site-cost scale                 | $\kappa^S$                   | 0.22                            | 2024 solar: 136 GW                |
| <i>C. Policy process (next slide)</i> |                              |                                 |                                   |
| Persistence on / off                  | $\pi_{11}, \pi_{00}$         | 0.90, 0.94                      | party control, 1993–2025          |

# Calibration: Policy Process

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- Every past lapse was short and renewed  $\Rightarrow$  history cannot tell how long a lapse lasts
- Going forward, the subsidy follows **which party controls government**: on under Democratic control of the presidency and Congress, off under Republican control
- Estimated from elections 1993–2025  $\Rightarrow$  a lapse lasts **about 17 years**; the subsidy is on **37%** of the time

► Scenarios

► Party control

► Parameters

# Results

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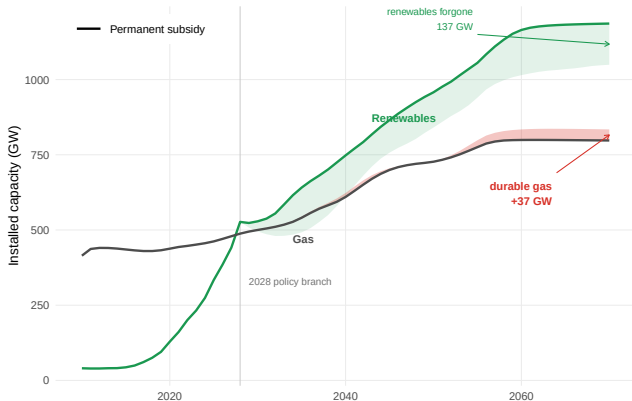
1. The value of permanence at a fixed budget
2. The transition across space
3. Permanence under perfect foresight

# Counterfactual Experiment

|                       | Permanent subsidy            | Uncertain subsidy         |
|-----------------------|------------------------------|---------------------------|
| 2010–2027             | same realized policy history |                           |
| From 2028             | always on                    | on/off with party control |
| Wind subsidy when on  | \$26/MWh                     | \$36.4/MWh                |
| Solar subsidy when on | 30% of capital cost          | 30% of capital cost       |
| Expected spending     | \$231B                       | \$231B                    |

- Same budget  $\Rightarrow$  continuous support vs. stochastic on-and-off support
- Solar rate held at 30%; the wind rate is scaled up to hold expected spending fixed
- **Value of permanence** = welfare under permanent – uncertain subsidy; the baseline comparison combines intermittency and uncertainty about policy dates
- Capacity measured at 2070; emissions summed 2010–2070

# Results 1: Capacity



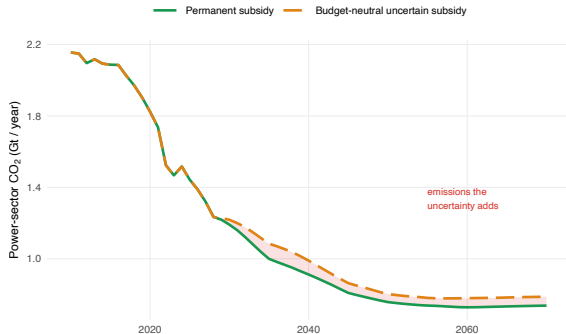
## Renewables: +137 GW at 2070

- 81 GW wind + 56 GW solar

## Gas: -37 GW

- Built during lapses by marginal regions
- Irreversible: stays and runs 30–40 years

# Results 1: Emissions



## Cumulative power-sector CO<sub>2</sub>, 2010–2070

|                   |                       |
|-------------------|-----------------------|
| Uncertain subsidy | 71.82 Gt              |
| Permanent subsidy | 69.53 Gt              |
| Gap               | <b>2.29 Gt (3.2%)</b> |

## Where the gap comes from

- 1.83 Gt: existing gas & coal run more while renewables are missing
- 0.46 Gt: the 37 GW of new gas

≈ 1.5 years of current U.S. power-sector emissions

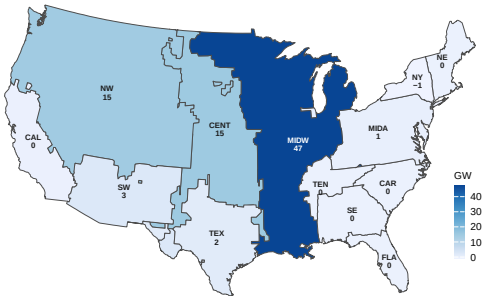
## Results 1: Welfare

| Permanent – uncertain subsidy      | \$B, PV     |
|------------------------------------|-------------|
| Avoided climate damage             | +321        |
| Consumption cost                   | –113        |
| Extra generation investment        | –251        |
| Higher renewable & baseload income | +144        |
| Lower gas & coal profit            | –23         |
| Higher wages                       | +16         |
| <b>Value of permanence</b>         | <b>+208</b> |

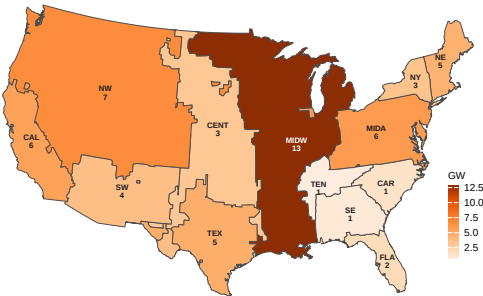
- Positive for any social cost of carbon above **\$67/t**
- Net cost is **0.13%** of consumption

# Results 2: Regional Capacity

(a) Additional wind capacity (GW)



(b) Additional solar capacity (GW)

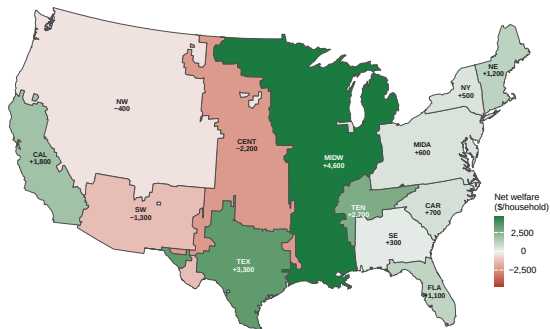


► Table    ► Over time

## Results 2: Regional Welfare

|            | Avoided damage | Consumption change | Net         |
|------------|----------------|--------------------|-------------|
| Midwest    | +115           | +0                 | <b>+115</b> |
| California | +48            | -17                | <b>+31</b>  |
| Texas      | +41            | -12                | <b>+29</b>  |
| Southwest  | +10            | -14                | -4          |
| Northwest  | +21            | -26                | -4          |
| Central    | +9             | -23                | -14         |
| National   | +321           | -113               | <b>+208</b> |

\$B, permanent — uncertain



Net gain per household

- Avoided damage is credited to the region whose **own plants** emit less
- **Midwest** builds and replaces its own fossil; **California** buys clean power; **Central, Northwest, Southwest** are net exporters of clean power and lose

## Results 3: Permanence Under Perfect Foresight

| Policy                         | First lapse<br>(years) | PV subsidy spending<br>(\$B) | Value of permanence<br>(\$B) |
|--------------------------------|------------------------|------------------------------|------------------------------|
| Permanent subsidy              | —                      | 230.9                        | —                            |
| Perfect foresight: short lapse | 6                      | 232.0                        | <b>245</b>                   |
| Perfect foresight: long lapse  | 17                     | 232.3                        | <b>365</b>                   |

- Firms learn the **entire future subsidy schedule in 2028**; there is no uncertainty about switch dates
- Known dates generate **investment bunching** around eligibility windows

**Predictability changes when firms invest, but it does not eliminate the value of continuous support.**

# Discussion

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| Omitted mechanism                                    | Effect on value of permanence | Why                                                                          |
|------------------------------------------------------|-------------------------------|------------------------------------------------------------------------------|
| Local pollution (SO <sub>2</sub> , NO <sub>x</sub> ) | ↑                             | Health benefits of less gas                                                  |
| Learning-by-doing                                    | ↑ (small)                     | Permanent subsidy learns faster; but learning is largely global              |
| Storage                                              | ambiguous                     | Raises renewable value: some regions build without subsidy; others lose more |
| Intraday solar/wind timing                           | ambiguous                     | Improves tech mix; residual still met by gas                                 |
| Renewable Portfolio Standard                         | ↓ (spatial)                   | Compresses regional results                                                  |

# Conclusion

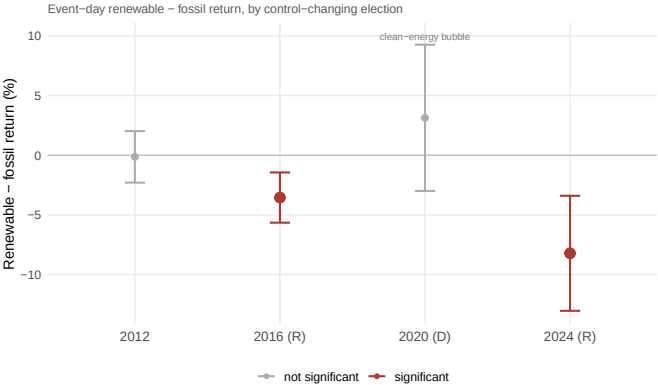
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- I develop a dynamic spatial general-equilibrium model with a detailed U.S. electricity sector
- I compute budget-neutral counterfactuals in which the subsidy switches on and off with the U.S. political cycle
- At the same budget, a permanent subsidy:
  - Adds 137 GW of renewables
  - Avoids 37 GW of irreversible gas
  - Avoids 2.29 Gt of CO<sub>2</sub>
  - Generates an aggregate welfare gain with a present value of \$208 billion
- Under perfect foresight, permanence is still valuable: \$245B with a known 6-year lapse and \$365B with a known 17-year lapse

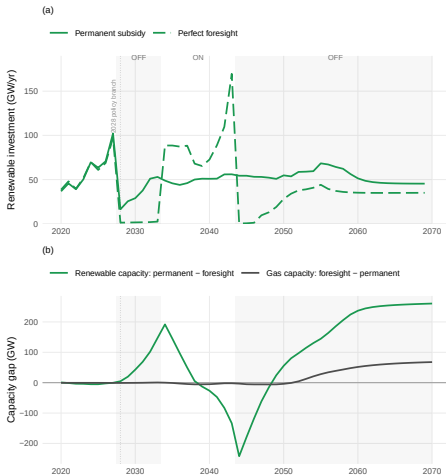
# Thank you!

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# Markets and Party Control, 2012–2024



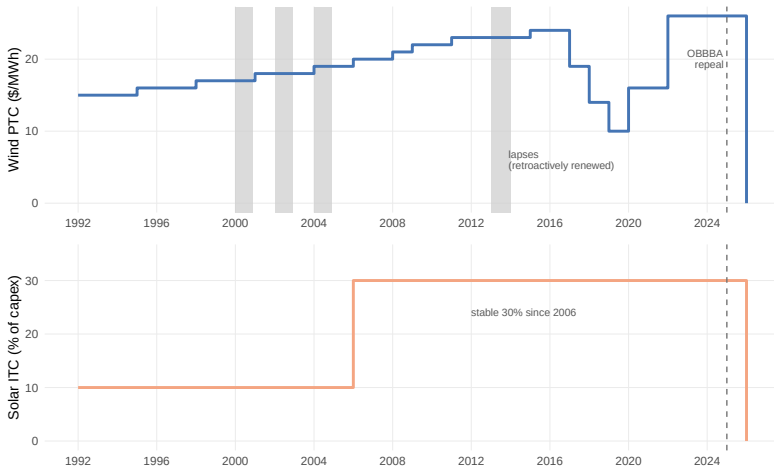
# Perfect Foresight: Investment Bunching



Short-lapse path: OFF 2028–2033, ON 2034–2043, OFF from 2044; the full schedule is known in 2028.

◀ Back

# History of the Wind and Solar Subsidies



# Intuition: A Subsidy Lapse Delays Renewable Investment

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## Renewables (wind, solar)

- High up-front, **irreversible** capital
- In many regions, profitable **only with the subsidy**
- A project started in a lapse never gets it

## Natural gas

- Dispatchable: the backup for demand peaks
- Value **independent** of the subsidy
- Also irreversible: runs 30–40 years

During the lapse, existing coal and gas generate more, and some additional gas is built as demand grows.

## Subsidy Vintage (Start-of-Construction Rule)

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$$K_{j,t+1}^{W,PTC} = (1 - \delta^W)(1 - h^{exit})K_{jt}^{W,PTC} + I_{jt}^W \cdot \mathbf{1}[s_t = 1], \quad h^{exit} = 0.1$$

- Wind built during a lapse adds to  $K^W$  but never earns the subsidy
- $h^{exit} = 0.1$  ages eligible vintages out over the 10-year window

# Model Fit

|                             | Moment                                      | Model | Observed |
|-----------------------------|---------------------------------------------|-------|----------|
| Aggregate                   | Power-sector CO <sub>2</sub> , 2010 (Gt/yr) | 2.15  | 2.2      |
|                             | Power-sector CO <sub>2</sub> , 2024 (Gt/yr) | 1.52  | 1.5      |
|                             | Wholesale price, 2021–24 (\$/MWh)           | 41    | 35–45    |
| Generation mix<br>2024 (%)  | Wind                                        | 10    | 10       |
|                             | Solar                                       | 8     | 5        |
|                             | Baseload (hydro+nuclear)                    | 25    | 24       |
|                             | Fossil (gas+coal)                           | 57    | 58       |
| Regional (Spearman $\rho$ ) | Wind capacity, 13 regions                   |       | 0.96     |
|                             | Solar capacity, 13 regions                  |       | 0.57     |
| Trade (EIA-930)             | Net-position sign match                     |       | 8/12     |
|                             | California net position (GW)                | –5.8  | import   |

Only national wind and solar capacity are targeted.

► Back

► Capacity paths

► Prices

# Comparison with Macro-Energy Models

|            | Standard macro-energy model | This model                                                         |
|------------|-----------------------------|--------------------------------------------------------------------|
| Technology | clean vs. dirty aggregate   | 6 technologies; <b>capacity <math>\neq</math> generation</b>       |
| Price      | one price or marginal cost  | set by the <b>last plant dispatched</b> ; falls as renewables grow |
| Demand     | smooth                      | <b>hourly peaks</b> (Pareto) $\Rightarrow$ backup capacity needed  |
| Space      | one economy                 | 13 regions; <b>trade within 3 interconnections</b>                 |
| Policy     | taken as given              | <b>stochastic</b> subsidy, per MWh (wind) and per MW (solar)       |

► Back

# Policy Process: Parameters

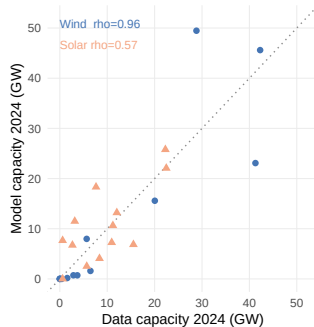
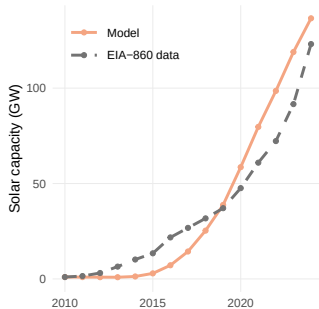
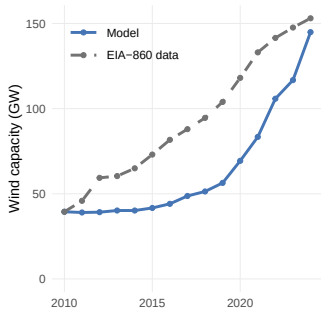
Subsidy **on**: wind \$26/MWh, solar 30%    Subsidy **off**: wind \$0, solar 0%

|                           | Stay on<br>$\pi_{11}$ | Stay off<br>$\pi_{00}$ | Years on<br>(avg.) | Years off<br>(avg.) | Share of<br>time on |
|---------------------------|-----------------------|------------------------|--------------------|---------------------|---------------------|
| Beliefs 2010–2024 (wind)  | 0.857                 | 0.70                   | 7                  | 3.3                 | 68%                 |
| <b>Central, from 2025</b> | <b>0.90</b>           | <b>0.94</b>            | <b>10</b>          | <b>17</b>           | <b>37%</b>          |
| Short lapses (robustness) | 0.90                  | 0.40                   | 10                 | 1.7                 | 85%                 |

2010–2024 beliefs: investors acted as if lapses would last (Barradale 2010). Split control leaves the subsidy unchanged.

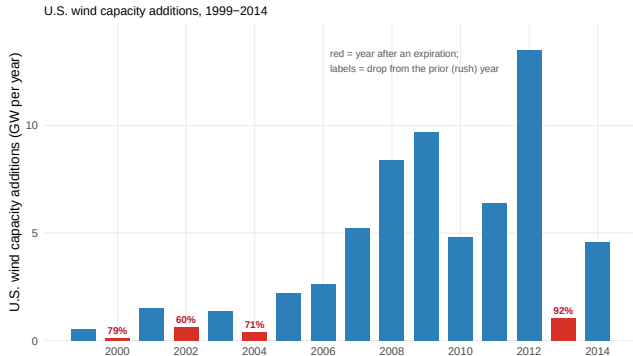
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# Model Fit: Capacity, 2010–2024



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# Wind Investment Around Subsidy Lapses



- Because the subsidy decides whether much of wind pays, developers time projects around it
- Additions surge before each deadline, then fall **79%, 60%, 71%, 92%**
- Solar, on a stable subsidy, grew smoothly [► Solar](#)
- Every lapse was short  $\Rightarrow$  a lasting repeal **has never happened before**

[► Back](#)[► Regressions](#)[► Barradale \(2010\)](#)[► PPA negotiations](#)

# Robustness

| Check            | Variation                           | Value of permanence | Breakeven SCC |
|------------------|-------------------------------------|---------------------|---------------|
| Reporting window | to 2050                             | \$85B               | \$109/t       |
|                  | <b>to 2070 (baseline)</b>           | <b>\$208B</b>       | <b>\$67/t</b> |
|                  | to 2090                             | \$208B              | \$67/t        |
| Forward regime   | short lapses (1.7 years on average) | \$43B               | \$21/t        |
|                  | permanent repeal                    | \$255B              | \$94/t        |

[▶ Back](#)[▶ SCC figure](#)[▶ All regimes](#)

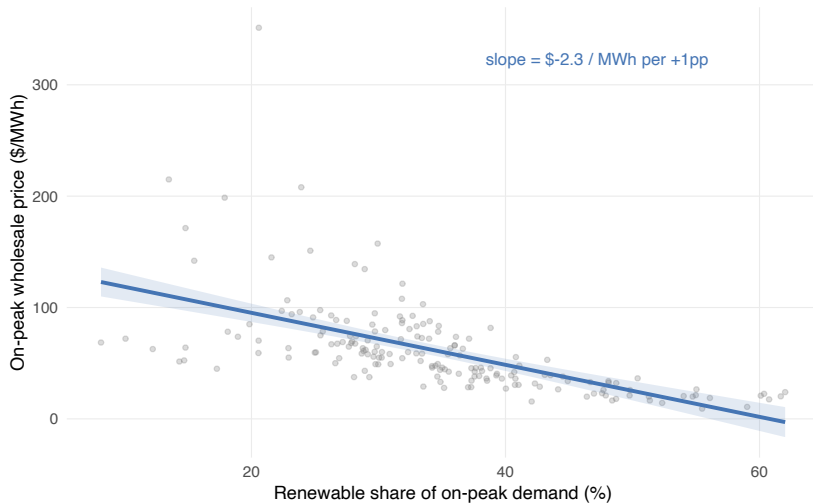
# Empirical Support for Model Assumptions

---

- **Prices fall with renewables:** +1 pp renewable share  $\Rightarrow$   $-\$1.4$  to  $-\$1.7/\text{MWh}$  on-peak  
▶ Figure
- **Gas plants last decades:** units run 40–60 years; median age 19  $\Rightarrow$  investment is irreversible  
▶ Figure
- **Three separate markets:** hub prices correlate 0.77 within vs. 0.20 across interconnections  
▶ Figure
- **Peakier demand, more gas:** regions with more variable load hold more gas ▶ Figure

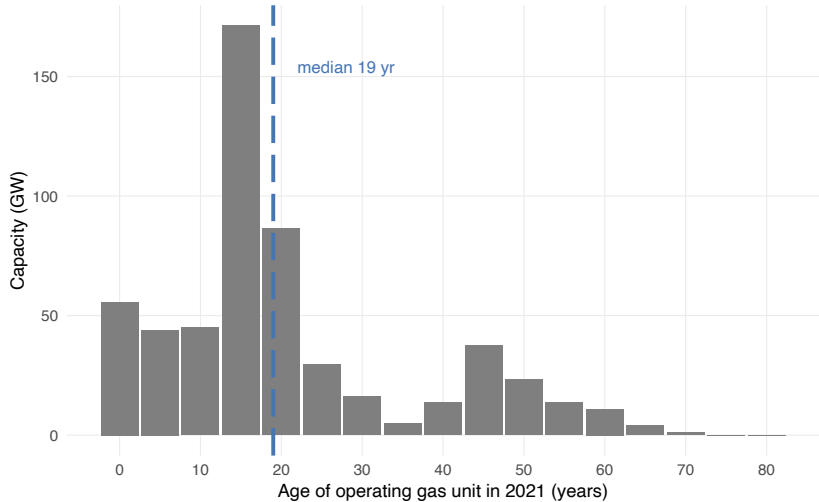
▶ Back

# Prices Fall with Renewables



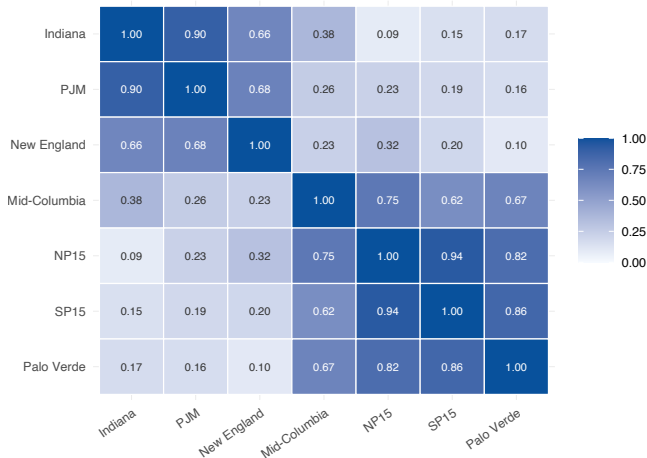
California, daily on-peak, 2023–2024. [▶ Back](#)

# Gas Plants Last Decades



Operating gas capacity by age, 2021. [▶ Back](#)

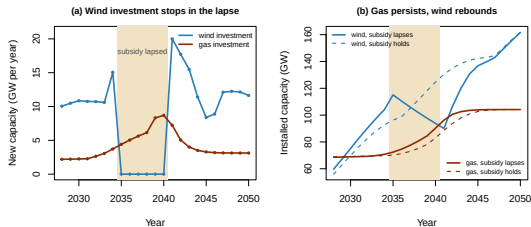
# Three Separate Markets



Correlation of daily on-peak prices across hubs, 2023–2024.

► Back

# The Full Model: One Lapse in the Midwest



## (a) Investment: one known lapse, 2035–2040

- Subsidy on:  $\sim 11$  GW/yr of wind
- Lapse: wind stops, gas rises to  $\sim 9$  GW/yr
- Subsidy back: wind rebounds to  $\sim 20$  GW/yr

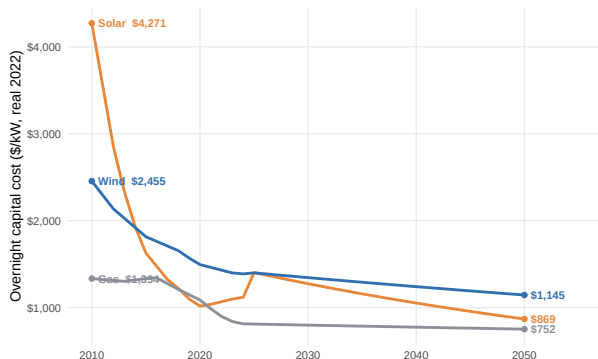
► Back

## (b) Capacity vs. no lapse

- Wind dips, then is rebuilt
- Gas built in the lapse **stays**
- Gas built during the lapse remains in the capital stock until it depreciates

► Wind vs. solar

# Capital Costs Fell; Wind Still Costs More



Overnight capital cost, 2010 → long run (\$/kW, real 2022):

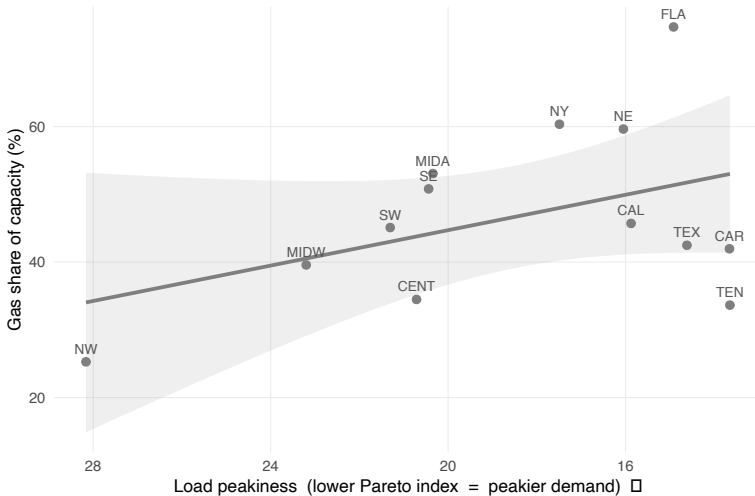
- Solar: 4,270 → 790
- Wind: 2,460 → 1,100
- Gas: 1,330 → 740

Cost per MW is not cost per MWh: wind runs 11–36% of the time, so many sites still need the subsidy to beat the wholesale price.

Sources: LBNL/EIA (history), NREL ATB 2024 Moderate (projection).

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# Peakier Demand, More Gas



Regions with more variable load hold more gas: renewables cannot cover the peaks.

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# Parameters

| Parameter                                                  | Symbol                         | Value                           | Source              |
|------------------------------------------------------------|--------------------------------|---------------------------------|---------------------|
| <i>Panel A. External</i>                                   |                                |                                 |                     |
| Discount factor                                            | $\beta$                        | 0.9615                          | $\rho \approx 4\%$  |
| Depreciation wind, solar, gas, $K_P$                       | $\delta$                       | 0.04, 0.033, 0.03, 0.05         | NREL ATB            |
| Wind / solar subsidy                                       | $\phi^{PTC}, \phi^{ITC}$       | \$26/MWh, 30%                   | \$45, \$48          |
| Capital costs (\$M/MW)                                     | $p_t^S, p_t^W, p_t^G$          | 4.27→0.79, 2.46→1.10, 1.33→0.74 | LBNL, ATB           |
| Demand elasticity                                          | $\sigma$                       | 0.3                             | lit. 0.2–0.4        |
| Emission factors (t/MWh)                                   | $\xi^G, \xi^C$                 | 0.40, 0.95                      | eGRID               |
| Social cost of carbon                                      | SCC                            | \$190/t                         | EPA 2023            |
| Site-cost curvature                                        | $\eta$                         | 2                               | NREL                |
| Capacity factors                                           | $CF_j^k$                       | wind 0.11–0.36, solar 0.17–0.27 | EIA                 |
| Cost shares                                                | $\alpha_K, \alpha_L, \alpha_E$ | 0.33, 0.65, 0.02                | BEA                 |
| <i>Panel B. Internal (match 2010–2024 capacity growth)</i> |                                |                                 |                     |
| Wind, solar adjustment cost                                | $\bar{\psi}^W, \bar{\psi}^S$   | $2 \times 10^5$                 | 143 / 136 GW (2024) |
| Gas adjustment cost                                        | $\bar{\psi}^G$                 | $10^7$                          |                     |
| Wind / solar site-cost scale                               | $\kappa^W, \kappa^S$           | 0.08 / 0.22                     |                     |

► Data

► Exogenous paths

# Welfare and Counterfactuals

Welfare (reported separately): ► Equilibrium

$$\mathcal{W} = \underbrace{\sum_t \beta^t \sum_j N_j \log U(C_{jt}/N_j, E_{jt}^{HH}/N_j)}_{\text{consumption}} - \underbrace{\text{SCC} \sum_t \varepsilon_t}_{\text{climate damage (SCC} \times \text{tons)}}$$

**Counterfactuals:** common 2010–2027 history; differ only in forward  $\Pi$  from 2028

- **Permanent:**  $s_t = 1$  (\$231B in present value)
- **Uncertain:** estimated switching process (on under Democratic control, off under Republican)
  - **Budget-neutral:** expected spending \$231B  $\Rightarrow$  wind \$36.4/MWh when active; solar 30%
  - Realistic post-OBBBA: \$161B
- **No subsidy:**  $s_t = 0$

**Value of permanence = Permanent – Budget-neutral uncertain**

# Event Study: Cumulative Abnormal Returns

Renewable vs. fossil CARs relative to the market, scaled by pre-event volatility; on impact  $[0, +3]$ , placebo  $[-5, -1]$

| Event                  | Date         | On impact $[0, +3]$ |      |            | Placebo $[-5, -1]$  |
|------------------------|--------------|---------------------|------|------------|---------------------|
|                        |              | Renew. (%)          | $t$  | Fossil (%) | Renew. – fossil $t$ |
| IRA deal announced     | Jul 28, 2022 | +12.8               | 2.6  | +0.8       | 0.3                 |
| 2024 general election  | Nov 6, 2024  | −20.7               | −4.8 | +0.7       | 0.7                 |
| OBBA House passage     | May 22, 2025 | −11.0               | −2.9 | −1.3       | −0.4                |
| OBBA Senate (softened) | Jul 1, 2025  | +9.9                | 2.1  | +1.2       | 2.1                 |

- Renewables move on the news; fossil stays within 1.3%
- Placebo insignificant for the IRA, the election and House passage ( $|t| \leq 0.7$ ); Senate revision  $t \approx 2$

# Regional Capacity Response, 2070

| Region       | $\Delta$ Wind | $\Delta$ Solar | $\Delta$ Total |
|--------------|---------------|----------------|----------------|
| Midwest      | 47.1          | 12.8           | 59.9           |
| Northwest    | 14.6          | 6.7            | 21.3           |
| Central      | 15.1          | 3.3            | 18.4           |
| Texas        | 1.9           | 4.7            | 6.7            |
| Southwest    | 2.8           | 3.7            | 6.5            |
| Mid-Atlantic | 0.5           | 5.9            | 6.4            |
| California   | 0.0           | 5.5            | 5.6            |
| New England  | 0.0           | 4.5            | 4.5            |
| New York     | -0.8          | 3.5            | 2.6            |
| Florida      | 0.0           | 2.0            | 2.0            |
| Carolinas    | 0.0           | 1.4            | 1.4            |
| Southeast    | 0.0           | 1.0            | 1.0            |
| Tennessee    | 0.0           | 0.7            | 0.7            |
| National     | 81            | 56             | 137            |

GW at 2070, permanent — budget-neutral uncertain

- Wind response concentrated in the **interior wind belt**: Midwest, Northwest, Central add **76 GW**
- Low-wind regions (Tennessee, Florida) build no wind even with a permanent subsidy
- **Texas**: isolated and wind-rich, rebuilds quickly when the subsidy returns  $\Rightarrow$  2070 gap only 1.9 GW
- Exporters: Midwest, Central, Northwest; importers: California, Mid-Atlantic, Florida

## Regional Welfare, All Regions

| Region       | Avoided<br>(\$B) | $\Delta C$<br>(\$B) | Net<br>(\$B) | Net/HH<br>(\$) |
|--------------|------------------|---------------------|--------------|----------------|
| Midwest      | +115             | +0                  | +115         | +4,600         |
| California   | +48              | -17                 | +31          | +1,800         |
| Texas        | +41              | -12                 | +29          | +3,300         |
| Mid-Atlantic | +26              | -9                  | +17          | +600           |
| Tennessee    | +4               | +8                  | +11          | +2,700         |
| New England  | +16              | -7                  | +9           | +1,200         |
| Florida      | +13              | -5                  | +8           | +1,100         |
| New York     | +10              | -5                  | +4           | +500           |
| Carolinas    | +6               | -2                  | +4           | +700           |
| Southeast    | +3               | -2                  | +2           | +300           |
| Southwest    | +10              | -14                 | -4           | -1,300         |
| Northwest    | +21              | -26                 | -4           | -400           |
| Central      | +9               | -23                 | -14          | -2,200         |
| National     | +321             | -113                | +208         | +1,500         |

- **Builders that displace own fossil** gain most: Midwest, Texas
- **Importers** gain from clean power they buy: California, Mid-Atlantic
- **High-cost exporters** lose: pay to build, avoided emissions accrue to consumers (Central -\$11B)
- Only CO<sub>2</sub> counted  $\Rightarrow$  regional benefits are a **lower bound** (no SO<sub>2</sub>/NO<sub>x</sub>)

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# Equilibrium and Welfare

**Within-period equilibrium:** given  $(\mathbf{K}_{jt}, s_t)$ , prices  $\{p_{jt}^E, \bar{p}_{mt}^E\}$ , dispatch, demands and net trades s.t. merit order, optimal demand, LMP basis, interconnection clearing (fixed point: price  $\leftrightarrow$  demand)

**Dynamic equilibrium:** investment solves the Bellman with irreversibility; SDF consistent with consumption; capital, labor, goods markets clear;  $s_t \sim \Pi$ ; converge to stochastic steady state [► Algorithm](#)

**Welfare** — two components reported separately:

$$\mathcal{W} = \underbrace{\sum_t \beta^t \sum_j N_j \log U(C_{jt}/N_j, E_{jt}^{HH}/N_j)}_{\mathcal{W}^C: \text{consumption (CE \%)}} - \underbrace{\sum_t (1.02)^{-(t-2010)} \text{SCC}_t \mathcal{E}_t}_{\mathcal{W}^E: \text{climate damage}}$$

- Climate damage discounted to the base year at the EPA's 2% social rate; consumption at the model's rate
- Subsidies, taxes, trade rebates are transfers; investment valued at full  $p^k$

# Firms and the Capital Market

---

**Final-good firm** (CES in a capital–labor bundle and electricity):

$$Y_{jt} = Z_{jt} \left[ (1 - \gamma_E)^{1/\sigma} (K_{P,jt}^{\hat{\alpha}_K} L_{jt}^{\hat{\alpha}_L})^{\frac{\sigma-1}{\sigma}} + \gamma_E^{1/\sigma} (E_{jt}^{firm})^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

- $\gamma_E$  set so electricity cost share = 2% (BEA)
- Same elasticity  $\sigma$  as households

**Electricity demand** (price-elastic):

$$E_{jt}^{firm} = \bar{\alpha}_E Y_{jt} (p_{jt}^E)^{-\sigma}, \quad E_{jt}^{HH} = \bar{\alpha}_h C_{jt} (p_{jt}^E)^{-\sigma}$$

**National capital market:** households save  $A_{jt}$ ; firms rent  $K_P$  at  $r_t^K$ ;  $\sum_j K_{P,jt} = \sum_j A_{jt}$

# Why a Separate Bellman? Fisher Separation

The household-generator does two jobs, and they separate:

- (i) **Run the generation business:** invest in  $K^S, K^W, K^G$
- (ii) **Split wealth:** choose consumption  $C$  vs. saving  $S$

Because it saves in the national capital market at a return  $r^K$  it takes as given, **Fisher separation** holds:

- Run generation to **maximize its market value**, discounted at the SDF  $M_{t,t+1} = \beta U'_C / U_C$  — *independent of the consumption path*
- Then allocate the resulting wealth between  $C$  and  $S$  (Euler equation)

**Why saving is not in the Bellman:** it is the frictionless, reversible margin (no  $I \geq 0$ , no policy state), pinned by the Euler equation. The Bellman is reserved for generation capital, which is **irreversible**, carries the policy state  $s_t$  and the vintage  $K^{W,PTC}$ , and has adjustment and site costs  $\Rightarrow$  option value.

Link:  $M$ , prices  $p^E$ , and  $r^K$  are solved to consistency in the outer GE loop.

# Is It GE? Are Markets Complete?

## Is it general equilibrium? **Yes.**

- Every price the region takes as given ( $p^E$ ,  $r^K$ ,  $M$ , capture prices  $q^W, q^S$ ) is **endogenous**, set by market clearing in the outer fixed point
- Clears: electricity (within each interconnection), the national capital market ( $\sum_j K_{P,jt} = \sum_j A_{jt}$ ), regional labor, and goods
- Price-taking  $\neq$  partial equilibrium: **cannibalization** (capture price falls as renewables enter) is the GE feedback that disciplines how much wind and solar get built

## Are markets complete? **Capital and goods yes; policy risk no.**

- There is no market that spans **regional policy risk**: the region values its own generation with its *own* SDF  $M_j$  (built from region  $j$ 's consumption)
- The outer loop reconciles the region-specific  $M_j$  with the common national  $r^K$

# Interregional Trade and Electricity Profit

---

**Trade within interconnection**  $m$ : net exports  $X_{jt} = G_{jt} - \bar{E}_{jt}$

$$p_{jt}^E = \bar{p}_{mt}^E - \gamma X_{jt} / TL_j$$

- Exporters paid below the hub, importers pay above
- Transmission cost  $\frac{\gamma}{2} X_{jt}^2 / TL_j$  is a real resource; the basis is a transfer rebated as  $R_{jt}$
- Market clearing:  $\sum_{j \in m} G_{jt} = \sum_{j \in m} \bar{E}_{jt}$

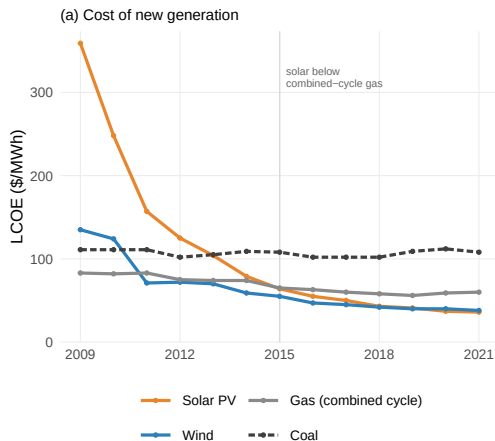
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# Data

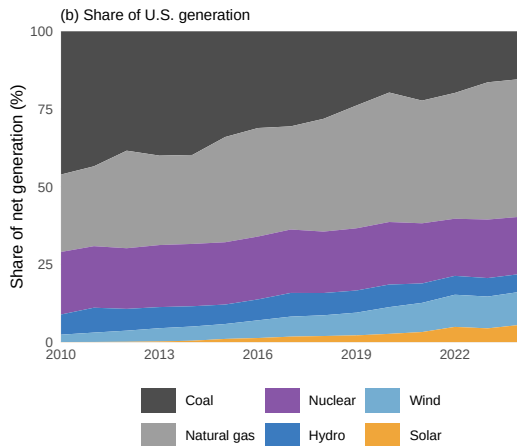
| Dataset                              | Years     | Use in the model                                |
|--------------------------------------|-----------|-------------------------------------------------|
| EIA-860 (capacity)                   | 2010–2024 | Initial stocks $K_{j,0}^k$ ; coal retirement    |
| EIA-923 (generation, fuel)           | 2010–2024 | Capacity factors $CF_j^k$ ; fuel costs          |
| EIA-930 (hourly demand, interchange) | 2021–2023 | Pareto shape $\theta_j$ ; trade capacity $TL_j$ |
| EIA AEO 2025 Reference               | 2024–2050 | Demand growth                                   |
| NREL ATB 2024 (Moderate)             | 2025–2050 | Renewable and gas cost decline                  |
| NREL developable resource            | 2023      | Site-cost scales $\bar{K}_j^k$                  |
| ICE, NYISO wholesale prices          | 2021–2025 | Trade slope $s_j$                               |
| EPA eGRID                            | 2022      | Emission factors $\xi^G, \xi^C$                 |
| U.S. Census                          | 2020      | Households $L_j$                                |
| PTC/ITC statutes, IRS notices        | 1992–2025 | Eligibility; Markov chain                       |

[▸ Definitions](#)[▸ Regional summary](#)

# The Transition Is Under Way



(a) Levelized cost by technology, 2009–2021 (Lazard).  
(EIA).



(b) Share of U.S. net generation, 2010–2024

# The Largest Reallocation of Capital

---

**“Transitioning the global economy from fossil fuels to sustainable sources of energy requires the largest reallocation of capital in human history”**

(Institute for Climate and Sustainability at Harvard University)

- The U.S. is the 2nd-largest CO<sub>2</sub> emitter; electricity is almost **a quarter** of U.S. emissions
- Wind and solar costs fell below fossil in the 2010s (wind  $\approx$ 2011, solar  $\approx$ 2013–15) ► LCOE
- After  $\sim$ 15 flat years, U.S. electricity demand is projected to grow **0.9–1.6%/yr through 2050** (data centers, electrification; EIA AEO 2025)

## Value of a Permanent Subsidy: All Regimes

| Permanent subsidy vs.            | Climate<br>(\$B) | Cons. cost<br>(\$B, PV) | Net<br>(\$B) | Breakeven SCC<br>(\$/t) |
|----------------------------------|------------------|-------------------------|--------------|-------------------------|
| Budget-neutral uncertain         | 321              | 113                     | +208         | 67                      |
| Realistic uncertain (post-OBBBA) | 340              | 135                     | +205         | 76                      |
| No subsidy                       | 502              | 247                     | +255         | 94                      |
| Cross-party                      | 48               | 5                       | +43          | 21                      |

| Forward regime (vs. budget-neutral) | $\Delta$ Emissions (%) | $\Delta$ W cons. (%) | $\Delta$ W incl. SCC (%) |
|-------------------------------------|------------------------|----------------------|--------------------------|
| Permanent                           | -3.19                  | -0.13                | +0.22                    |
| Realistic uncertain                 | +0.19                  | +0.04                | +0.02                    |
| No subsidy                          | +1.83                  | +0.18                | -0.01                    |
| Cross-party                         | -2.71                  | -0.10                | +0.19                    |

# Welfare Decomposition by Channel

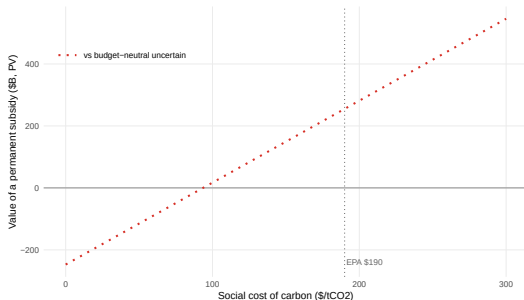
|                            | Spending<br>Uncertain—Matched | Permanence<br>Matched—Permanent | Total<br>Uncertain—Permanent |
|----------------------------|-------------------------------|---------------------------------|------------------------------|
| Wages                      | +3                            | −16                             | −13                          |
| Capital income             | +0                            | +0                              | +0                           |
| Renewable & baseload rents | −10                           | −144                            | −154                         |
| Gas & coal profit          | −2                            | +23                             | +21                          |
| Investment                 | +30                           | +251                            | +281                         |
| Net consumption cost       | +22                           | +113                            | +135                         |
| Climate (avoided damage)   | +19                           | +321                            | +340                         |
| <b>Value of commitment</b> | −3                            | <b>+208</b>                     | <b>+205</b>                  |
| Memo: subsidy transfer     | −80                           | +10                             | −70                          |

**Resource inefficiency of uncertainty (\$B):** excess adjustment +15, excess gas +11, premature deployment −24 ⇒ net +2: second-order. The true cost is the emissions of long-lived gas.

# Reporting Window and the SCC

| Window      | Gap (Gt)    | Value (\$B) | Breakeven   |
|-------------|-------------|-------------|-------------|
| 2050        | 1.30        | 85          | \$109       |
| <b>2070</b> | <b>2.29</b> | <b>208</b>  | <b>\$67</b> |
| 2090        | 3.21        | 208         | \$67        |

- Long-lived gas  $\Rightarrow$  the physical gap grows with the window (5.0 Gt by 2130), but its discounted value converges by 2070
- Consumption cost is front-loaded, nearly window-independent
- 2070 and 2090 give the same value: the horizon is not a knife-edge choice



## Single-Party Control Transitions, 1993–2025

| from \ to  | D trifecta | R trifecta | Divided |
|------------|------------|------------|---------|
| D trifecta | 0.00       | 0.00       | 1.00    |
| R trifecta | 0.00       | 0.33       | 0.67    |
| Divided    | 0.20       | 0.30       | 0.50    |

- Per two-year Congress. Ergodic:  $\Pr(D) = 0.12$ ,  $\Pr(R) = 0.27$ ,  $\Pr(X) = 0.61$
- Subsidy on after D trifecta, off after R, held under divided government
- From the 2025 R trifecta: long-run on-probability  $\approx 0.37$
- Annualized by matching spell lengths:  $\pi_{11} = 0.90$ ,  $\pi_{00} = 0.94$

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# The Wind PTC: Legislative Record

| Act                               | Enacted  | Horizon (yr) | Coalition | Senate maj. | Days lapsed |
|-----------------------------------|----------|--------------|-----------|-------------|-------------|
| Energy Policy Act                 | Oct 1992 | 6.5          | Divided   | D, 4        | —           |
| Ticket to Work Act                | Dec 1999 | 2            | Divided   | R, 5        | 170         |
| Job Creation Act                  | Mar 2002 | 2            | Divided   | D, 9        | 68          |
| Working Families Tax Relief Act   | Oct 2004 | 1            | Unified R | R, 9        | 278         |
| Energy Policy Act                 | Aug 2005 | 2            | Unified R | R, 5        | 0           |
| Tax Relief & Health Care Act      | Dec 2006 | 1            | Unified R | R, 5        | 0           |
| Emergency Econ. Stabilization Act | Oct 2008 | 1            | Divided   | D, 9        | 0           |
| Recovery Act (ARRA)               | Feb 2009 | 3            | Unified D | D, 2        | 0           |
| American Taxpayer Relief Act      | Jan 2013 | 1            | Divided   | D, 7        | 2           |
| Tax Increase Prevention Act       | Dec 2014 | 1            | Divided   | D, 5        | 353         |
| PATH Act (phase-down)             | Dec 2015 | 5            | Divided   | R, 6        | 352         |
| Inflation Reduction Act           | Aug 2022 | 10           | Unified D | D, 10       | 228         |

Senate maj.: majority party, seats short of 60. Source: CRS R43453.

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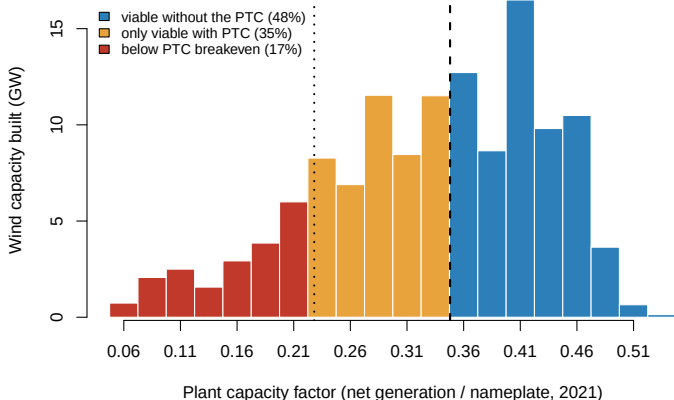
# A PTC Lapse Reduces Wind Investment

|                                              | (1)<br>Region panel | (2)<br>Rush + bust       | (3)<br>Next-year $\Delta$ | (4)<br>Wind vs. solar |
|----------------------------------------------|---------------------|--------------------------|---------------------------|-----------------------|
| PTC lapse                                    | −1.65***<br>(0.23)  | −1.44***<br>(0.27)       | −1.98***<br>(0.42)        | −1.12*<br>(0.64)      |
| [implied % change]<br>PTC expiry (rush year) | −81                 | −76<br>0.62***<br>(0.21) | −86                       | −67                   |
| Estimator                                    | PPML                | PPML                     | OLS                       | PPML                  |
| Smooth trend, region FE                      | ✓                   | ✓                        |                           |                       |
| Region $\times$ tech, year FE                |                     |                          |                           | ✓                     |
| Sample                                       | 1999–2024           | 1999–2024                | 2000–2024                 | 2011–2024             |
| Randomization-inference $p$                  | 0.000               |                          |                           |                       |
| Observations                                 | 262                 | 262                      | 25                        | 311                   |

$$\mathbb{E}[A_{jt}] = \exp(\gamma \text{Rush}_t + \beta \text{Lapse}_t + s(t) + \mu_j)$$

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## Fact 1: Capacity-Factor Distribution



Installed wind capacity by realized CF; dashed = merchant breakeven (0.35), dotted = with the PTC (0.23). At the \$50/MWh on-peak average, 35% is viable only with the credit. Sources: EIA-860, EIA-923.

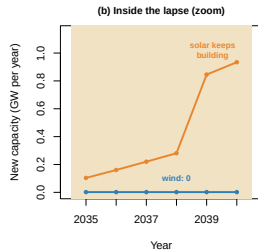
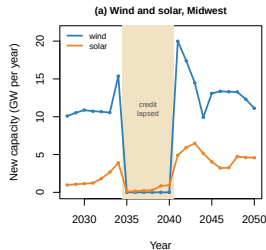
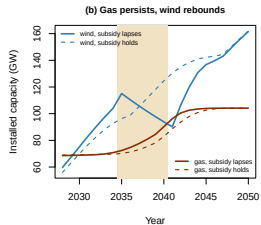
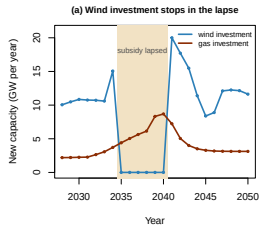
# The PTC in Wind Project Economics

Table: The production tax credit in the economics of a representative wind project.

|                                                           | \$/MWh |
|-----------------------------------------------------------|--------|
| <i>Revenue per MWh, during the ten-year credit period</i> |        |
| Wholesale energy price                                    | 50     |
| Production tax credit (\$45)                              | 26     |
| Total                                                     | 76     |
| <i>Levelized cost per MWh, unsubsidized</i>               |        |
| Capital (\$1,600/kW, 25 yr, 5%)                           | 39     |
| Fixed O&M (\$40/kW-yr)                                    | 14     |
| Levelized cost of energy                                  | 53     |
| <i>Net profit per MWh</i>                                 |        |
| Without the credit                                        | −3     |
| With the credit                                           | +23    |
| Credit as a share of revenue                              | 34%    |

Notes. All entries are dollars per megawatt-hour generated.

# Example: The Midwest Across a Lapse

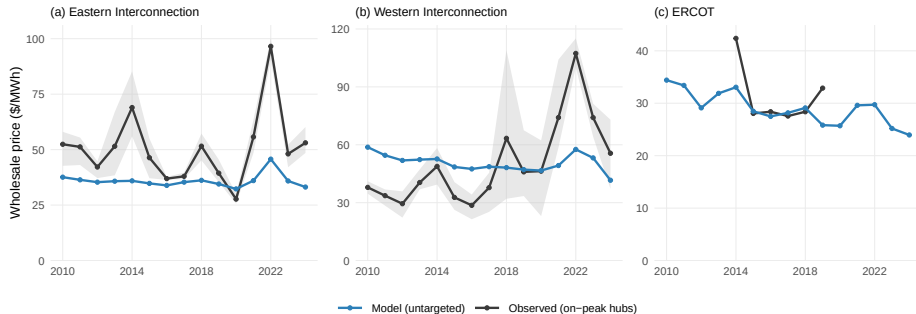


# Solution Method

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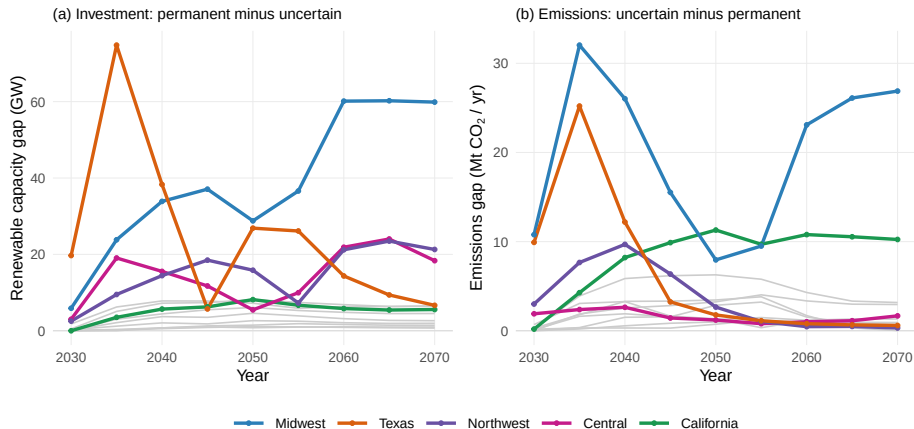
1. **Inner loop (within period):** given  $(\mathbf{K}_{jt}, s_t)$ , solve the fixed point between merit-order price and price-elastic demand, with interconnection clearing and LMP basis
2. **Backward induction:** from the stochastic steady state, solve the Bellman with irreversibility (KKT) for each region and policy state, given guessed price and SDF paths
3. **Forward simulation:** from the 2010 stocks, along the realized 2010–2027 policy history, then the forward  $\Pi$
4. **Outer loop:** update prices,  $r_t^K$ , transfers and SDFs until convergence

# Untargeted Wholesale Price, by Interconnection

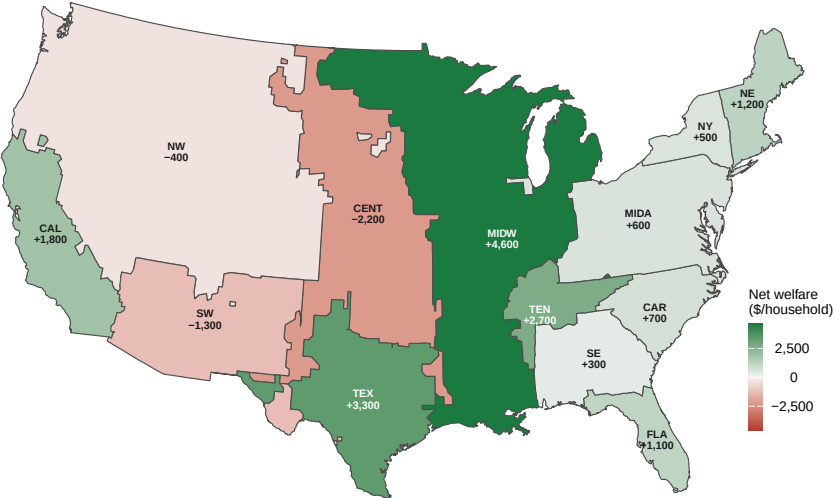


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# The Regional Gap Over Time

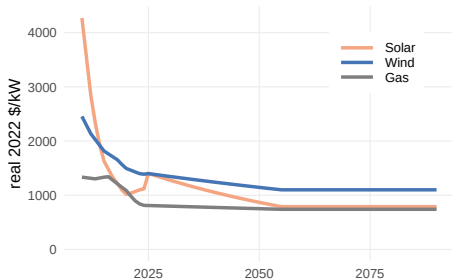


# Net Welfare per Household, by Region

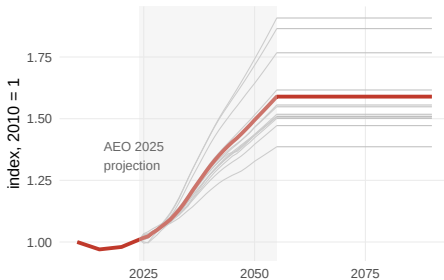


# Exogenous Processes

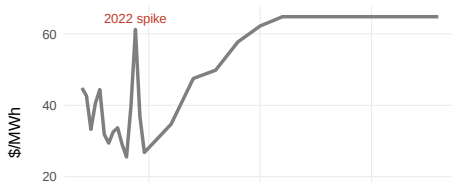
(a) Overnight capital cost



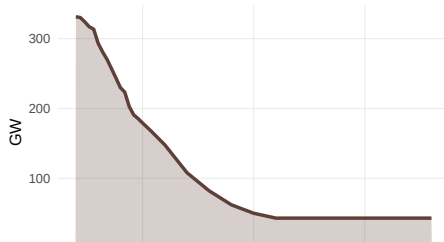
(b) Electricity demand (AEO 2025 Ref.)



(c) Gas marginal cost



(d) Coal capacity (retirement)



# Wind, Solar, and Gas by Region

Table: Summary statistics: the U.S. wind, solar, and gas sector by region.

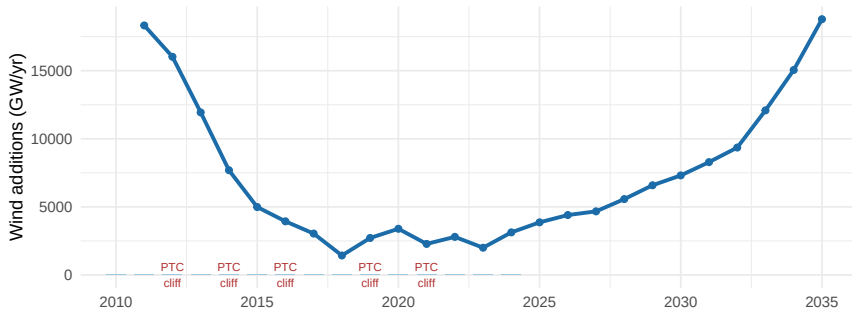
| Region       | Inter-connect | Wind CF | Solar CF | Wind (GW) | Solar (GW) | Gas (GW) | Demand (TWh) |
|--------------|---------------|---------|----------|-----------|------------|----------|--------------|
| Southwest    | West          | 0.38    | 0.32     | 5.7       | 7.6        | 20.7     | 117          |
| Central      | East          | 0.37    | 0.26     | 28.8      | 0.6        | 24.8     | 263          |
| Midwest      | East          | 0.34    | 0.21     | 41.3      | 15.5       | 98.0     | 667          |
| Texas        | ERCOT         | 0.34    | 0.29     | 42.3      | 22.5       | 64.9     | 403          |
| Northwest    | West          | 0.34    | 0.29     | 20.0      | 12.0       | 29.9     | 347          |
| Carolinas    | East          | 0.28    | 0.25     | 0.4       | 8.4        | 25.6     | 214          |
| California   | West          | 0.27    | 0.31     | 6.5       | 22.3       | 34.6     | 249          |
| Mid-Atlantic | East          | 0.27    | 0.19     | 3.7       | 11.2       | 90.3     | 751          |
| New England  | East          | 0.26    | 0.19     | 1.6       | 3.1        | 15.8     | 112          |
| New York     | East          | 0.23    | 0.19     | 2.9       | 2.7        | 20.7     | 140          |
| Tennessee    | East          | 0.10    | 0.25     | 0.0       | 0.6        | 6.9      | 186          |
| Southeast    | East          | 0.04    | 0.27     | 0.0       | 5.7        | 35.3     | 227          |
| Florida      | East          | 0.04    | 0.27     | 0.0       | 10.9       | 54.6     | 241          |
| U.S. total   |               |         |          | 153       | 123        | 522      | 3,919        |

Notes. Capacity factors are net generation over nameplate capacity, 2019–2024 average (EIA-923/EIA-860); regions with negligible wind show 0. Installed capacity is 2024 (EIA-860). Demand is 2024 electricity sales (AEO 2025). C EIA-860 EIA-923 EIA-860 AEO 2025

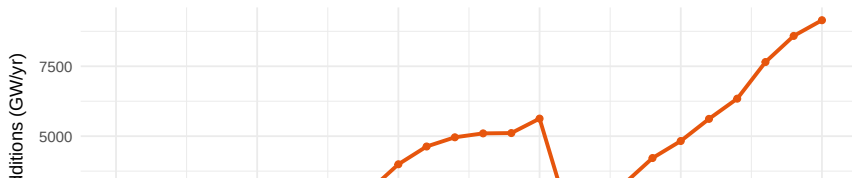
# Wind and Solar Additions

Model replicates the wind boom–bust around PTC cliffs

Bars = EIA-860 data; line = model baseline (mean path); dashed = PTC statutory expirations

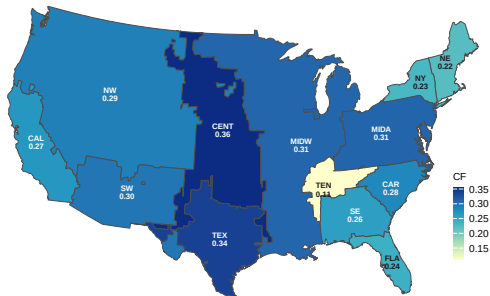


Solar (stable ITC): smooth in both data and model -- the boom–bust is wind-specific

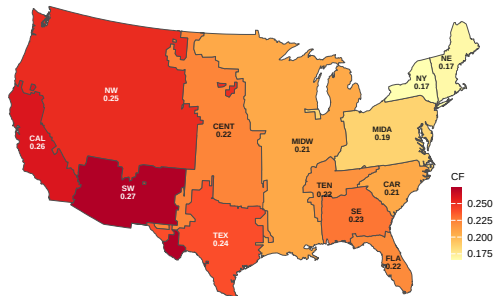


# Wind and Solar Capacity Factors by Region

(a) Wind capacity factor  $CF_j^W$



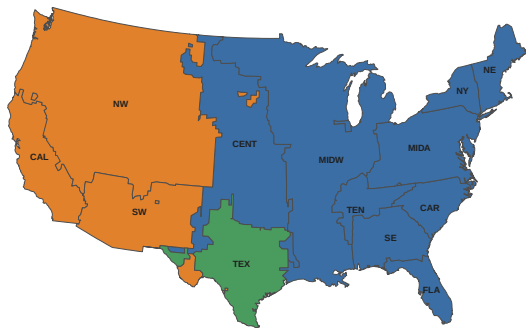
(b) Solar capacity factor  $CF_j^S$



Net generation (EIA-923) over nameplate (EIA-860), 2019–2024.

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# EIA Regions and Interconnections



Interconnection ■ Eastern ■ Western ■ Texas (ERCOT)

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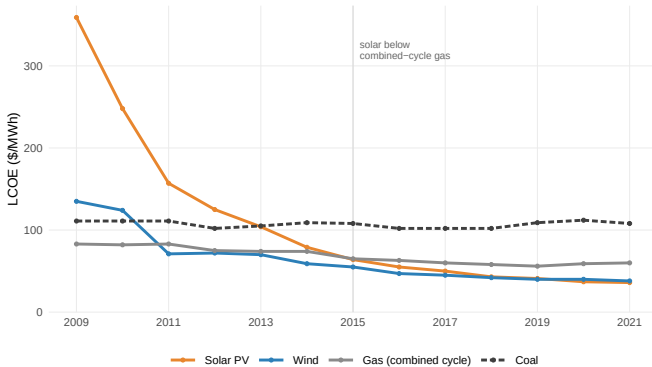
**Western:** Northwest, Southwest, California

**Eastern:** Florida, Southeast, Carolinas, Central, Midwest, Tennessee, New York, Mid-Atlantic, New England

**Texas (ERCOT):** independent, no interregional trade

Trade within interconnection at a congestion basis; none across.

# Renewables Are Cheaper per MWh (Levelized Cost)



# Negative Prices

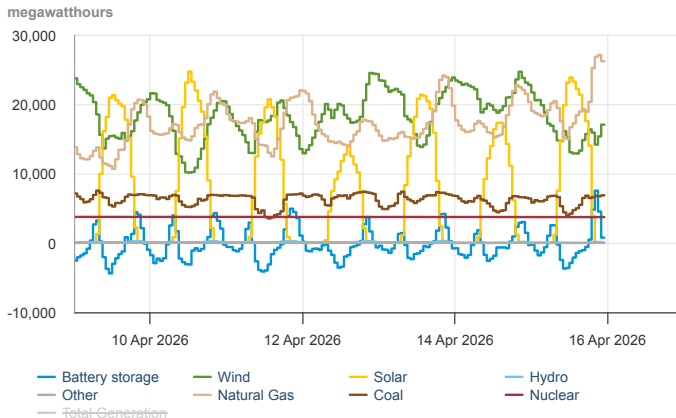
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# Texas Generation

Electric Reliability Council of Texas, Inc. (ERCOT) electricity generation by energy source 4/9/2026 – 4/16/2026, Central Time

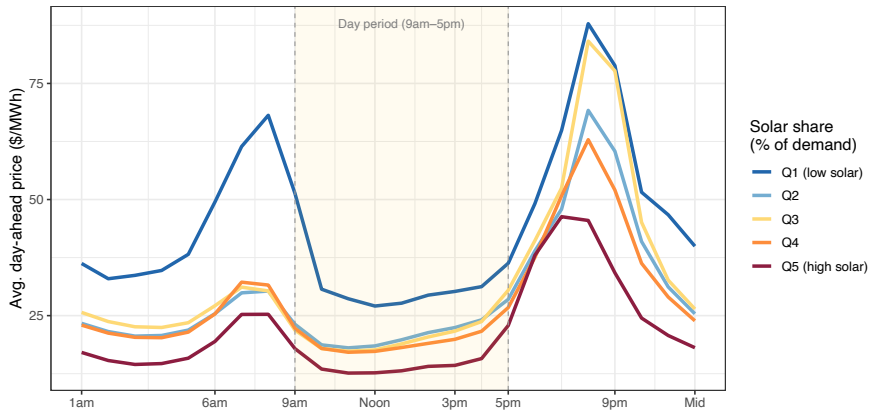


Data source: U.S. Energy Information Administration

# Duck Curve Price

## Higher Solar Penetration Deepens the Duck Curve

Avg. hourly DAM price by daily solar generation quintile — Texas (ERCOT)



Source: EIA-930 + ERCOT DAM NP4-190-CD (HB\_NORTH). Dec 2023 – present.

# Overnight Costs and Capacity Factors

## Overnight Capital Cost

Total cost a developer would incur during construction of a project, **excluding financing costs**.

### Components:

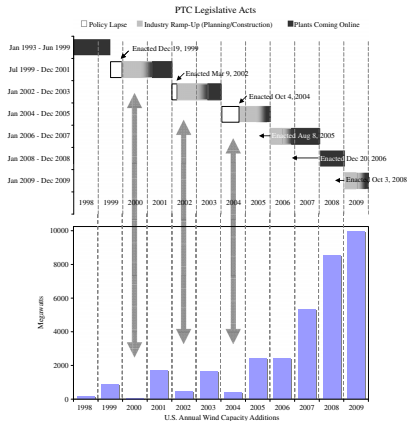
- Civil & structural materials and installation
- Mechanical equipment supply and installation
- Electrical, instrumentation, and controls
- Project indirect costs (engineering, construction management, start-up, commissioning, contractor overhead and profit)

Source: EIA (via Sargent & Lundy)

## Capacity Factor

The ratio of actual electrical energy produced by a generating unit to the electrical energy that could have been produced at continuous full power operation during the same period.

# Boom-Bust Cycle



**Fig. 1.** Even though PTC renewal has always been made retroactive to cover any policy lapses (white), investment stops while the industry waits for renewal. After the PTC is re-enacted, ramp-up time (gray) is required before new capacity can be brought online. Ramp-up time varies and is estimated to be 8–12 months following each enactment date. Even when the PTC is renewed prior to expiration, if re-enactment comes too late in the cycle, there may be a lull in investment while the industry ramps up again. The combination of policy lapses and the need for ramp-up time results in relatively short investment windows (black). Years in which there is no black (2000, 2002, and 2004) see sharp downturns in investment.

Source for PTC enactment dates: Wiser (2007), ENS (2008). Source for wind capacity additions: AWEA (2009).

Figure: Source: Barradale 2010

# PPA Negotiations

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PTC uncertainty is a barrier to transition:

- **Asymmetric assumptions stall negotiations.** Under PTC renewal uncertainty, IPPs (dependent on tax investors) must assume *no renewal* when pricing; utilities (obligated to secure least-cost power) must assume *renewal*. No overlap in bids  $\Rightarrow$  negotiations fail.
- **Retroactivity doesn't solve the problem.** Even though Congress has always renewed the PTC retroactively, this only helps single-party decisions. Multi-party PPA negotiations still collapse because utilities wait to avoid handing IPPs a windfall gain.
- **Not a pricing problem — a coordination problem.** Barradale (2010): PPA share drops from 97–98% in PTC “on” years to 51% in 2004 (off-year).

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